

# Amitava Dutta

PhD Scholar (*ongoing*)

Computational Geodynamics Lab, ECS, IISER Pune

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## Education

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|-------------|---|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 –      | 📖 | <b>PhD in Geophysics</b><br>Earth and Climate Sciences, IISER Pune, India.<br>Supervisor: Dr. Ajay Kumar                                       |
| 2019 – 2024 | 📖 | <b>Integrated BS-MS in Geological Sciences</b><br>Department of Earth Sciences, IISER Kolkata, India.<br>MS Project Advisor: Dr. Gaurav Shukla |

## Grants & Awards

[1 ] Awarded the CSIR-JRF in Earth Sciences to be funded by the Ministry of Earth Sciences.

## Research

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| 2024 – 2025 | 📖 | <b>Research Assistant</b><br>Computational Geodynamics Lab, IISER Pune, India.<br>Topic: Thermochemical evolution of continental lithosphere and its relation to mantle cooling: plate-tectonics and beyond.<br>Advisor: Dr. Ajay Kumar             |
| 2023 – 2024 | 📖 | <b>MS project</b><br>Computational Mineral Physics Lab, IISER Kolkata, India.<br>Topic: Effect of Hydration on the Iron Spin Crossover in Bridgmanite using <i>ab initio</i> DFT calculations.<br>Advisor: Dr. Gaurav Shukla                        |
| 2022 – 2023 | 📖 | <b>Semester Project</b><br>Topic: Crustal evolution of different Archean cratons using receiver functions analysis implementing H-K stacking and Joint Inversion.<br>Seismic Tomography Lab, IISER Kolkata, India.<br>Advisor: Dr. Kajaljyoti Borah |
| 2021        | 📖 | <b>Intern</b><br>GNSS Lab, Department of Physics, The University of Burdwan, India<br>Topic: Comparison of GNSS Precise Point Positioning (PPP) using various methods.<br>Advisor: Dr. Anindya Bose                                                 |
| 2019        | 📖 | <b>Intern</b><br>GNSS Lab, Department of Physics, The University of Burdwan, India<br>Topic: Studies on GNSS Pseudorange and Carrier Phase Residuals for Data Quality and Antenna Location Assessment.<br>Advisor: Dr. Anindya Bose                 |

## Research Publications

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### Conference Proceedings

- [1 ] A. Santra, **A. Dutta**, P. Banerjee and A. Bose, “A Study on Multi-GNSS Precise Point Positioning from India,” *URSI GASS 2020, Rome, Italy, August 2020*.

## Teaching

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Spring 2024



**Teaching Assistant**

Evolution of Earth & Life (EC1213), IISER Pune.

## Skills

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Programming & Scripting



Python, Fortran, C/C++, JAVA, Matlab, Bash Shell, Gnuplot, L<sup>A</sup>T<sub>E</sub>X

Geophysics & Geodynamics



Perple\_X, LitMod, Paraview, Seismic Analysis Code (SAC), Generic Mapping Tool (GMT), Quantum Espresso, Vesta

Operating System



Linux, Windows

Languages



Bengali, English, Hindi

## Workshops and Seminars

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May, 2025



GIAN: Plate Tectonics, Orogenic Cycle and Crustal Deformation, conducted by Interdisciplinary School of Science, Savitribai Phule Pune University, India. Course Instructors: Prof. Olivier Vanderhaeghe (GET, Université Paul Sabatier, Toulouse) and Dr. Mallika Jonnalagadda (SPPU). [Certificate]

April, 2024



Department of Earth Sciences (DES) day poster titled "Effect of Hydration on the Iron Spin Crossover in Bridgmanite". [Poster]

July, 2020



Satellite Photogrammetry and its Applications, conducted by Dr. Anil Kumar and Dr. Hima Pandey of the Indian Institute of Remote Sensing (IIRS). [Certificate]

July, 2020



Application of Geoinformatics in Ecological Studies, conducted by Dr. Hiten-dra Padalia of the Indian Institute of Remote Sensing (IIRS).

## Extra Curriculars

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**Football** - Love playing and have played in and organized multiple events.



**Photography** - Love Still Life and Nature Photography.



**Outreach** - Active member of the department outreach at IISER Kolkata from 2021 to 2023. [Certificate]

**References** : Please write me